

Qizhe (Charles) Yang

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EDUCATION

University of Southern California (USC) Sep 2023 – Jun 2027 (Expected)
B.S. in Computer Science; Minor in 3-D Animation in Cinematic Arts CA, USA

GPA: 3.72 / 4.00

Core Courses: Computer Graphics, Computer Systems, Software Engineering, Algorithms and Theory of Computing, Linear Algebra, Calculus, Probability Theory

Honors & Awards: Dean's List (2023–2024; 2024–2025); SOAR Scholarship

TECHNICAL SKILLS

Programming Languages: C++, Python (data science & ML libraries), Java, C, Assembly, TypeScript, JavaScript

Web & Frameworks: React, Flask, HTML, CSS

Tools & Platforms: Git/GitHub, Docker, ArcGIS Pro, L^AT_EX, Arduino, Teensy, Autodesk Maya, Final Cut Pro

Core Knowledge: Data Structures, Algorithms, Computer Systems, Computer Graphics

RESEARCH EXPERIENCE

SWAN Lab, University of Southern California Sep 2024 – Dec 2025
Undergraduate Research Assistant | Supervisor: Prof. Massoud Pirbazari

- Conducted research on PFAS contamination and water treatment strategies, focusing on system-level analysis of environmental remediation approaches
- Synthesized large-scale literature and domain data on PFAS properties, toxicity, detection methods, and removal technologies into a structured analytical framework
- Compared multiple PFAS mitigation technologies (e.g., activated carbon, anion exchange, membrane filtration) by modeling trade-offs in efficiency, scalability, cost, and waste generation
- Organized findings into modular, application-oriented summaries to support data-driven evaluation of current solutions and identify directions for future research

INTERNSHIP EXPERIENCE

BMW International Jan 2026 – Present
Assistant Artificial Intelligence Research Intern

- Conduct research on integrating Large Language Models (LLMs) into BMW's in-vehicle intelligent assistant system to enhance natural language understanding and interaction quality
- Benchmarked Alibaba Qwen and GPT-4o across 10 driving scenarios (100+ colloquial commands), evaluating intent classification accuracy and robustness under a 5-level simulated noise degradation framework
- Performed competitive analysis of intelligent voice ecosystems across BMW, Mercedes-Benz, Tesla, NIO, and XPeng, identifying optimization directions in emotional engagement, RAG-based driver manual retrieval, and wake-word/service integration
- Designed and implemented a multilingual voice assistant prototype in Flutter, integrating real-time weather APIs and LLM-powered dialogue processing (Chinese, English, German)
- Authored technical reports on Neue Klasse computing architecture, in-vehicle AI safety boundary design aligned with EU AI Act compliance, and intent recognition optimization strategies for ambiguous user commands

The Huntington Libraries, Art Galleries, and Botanical Gardens May 2025 – Jul 2025
GIS and Software Analyst Intern CA, USA

- Developed a web-based pathfinding system that computes shade-optimized walking routes using A* shortest-path algorithms
- Modeled urban shading data using computer-vision-processed inputs and integrated it into a graph-based routing framework

- Deployed the system as a public-facing web application to enable real-world validation of algorithmic results
- Demo & Code: walkhuntington.yangqizhe.com · Github: github.com/qizheYang/ShadowWalkerTheHuntington

PROJECTS

Ray Tracing & Monte Carlo Rendering

Sep 2024 – Present

USC CSCI 420 Course Project

- Implemented a basic ray tracing pipeline, including lighting, reflection, and shading models, as part of CSCI 420 assignments
- Explored Monte Carlo sampling techniques beyond course requirements to improve rendering realism

Travel Planner Web Application

Sep 2024 – Present

USC CSCI 201 Course Project

- Led a team-based full-stack project combining a React frontend with a Java backend using servlets and MySQL
- Implemented shared workspaces and live polls to support collaborative decision-making
- Integrated weather, hotel, and flight APIs for real-time travel planning
- Implemented backend concurrency and session management using Java servlets and multithreading to support real-time collaboration
- Code: github.com/qizheYang/usc_csci201_final_project

Personal Website

Sep 2024 – Present

- Designed and deployed a personal website using Vite, React, and TypeScript with a customized domain hosted on GitHub

Latin–Chinese Historical Text Translation

Sep 2024 – Present

- Collaborated in a multi-author translation project of classical Chinese philosophical texts by Martino Martini, segmenting long-form documents into aligned passages
- Designed and applied a structured annotation and revision workflow, explicitly flagging ambiguous terms and translation uncertainties for iterative review
- Coordinated version control across contributors by tracking edits, commentary notes, and translation rationale to ensure cross-language semantic consistency

Some Other for Fun Projects

Present

- Please visit rehydratedwater.com to see them.

EXTRACURRICULAR ACTIVITIES

USC Formula SAE Electric Vehicle Team

Sep 2024 – Dec 2025

Electrical & Communications Systems Engineer (Student Technical Team)

- Maintained and enhanced a GEVCU- and Teensy-based electric vehicle control system, integrating throttle input, motor control, Orion Battery Management System (BMS), and CAN bus communication
- Designed and implemented a driver dashboard interface to support real-time vehicle monitoring and system feedback during testing and competition
- Debugged and refactored legacy C++ / Arduino code to improve system reliability and hardware compatibility
- Developed and executed end-to-end test plans covering motor systems, BMS, CAN communication, and cooling controllers
- Developed and integrated a real-time dashboard system for Formula SAE Electric vehicles, interfacing with CAN bus telemetry for live system monitoring under competition constraints

ADDITIONAL INFORMATION

Languages: English (Fluent), Mandarin (Native)

Interests: Mahjong, Soccer